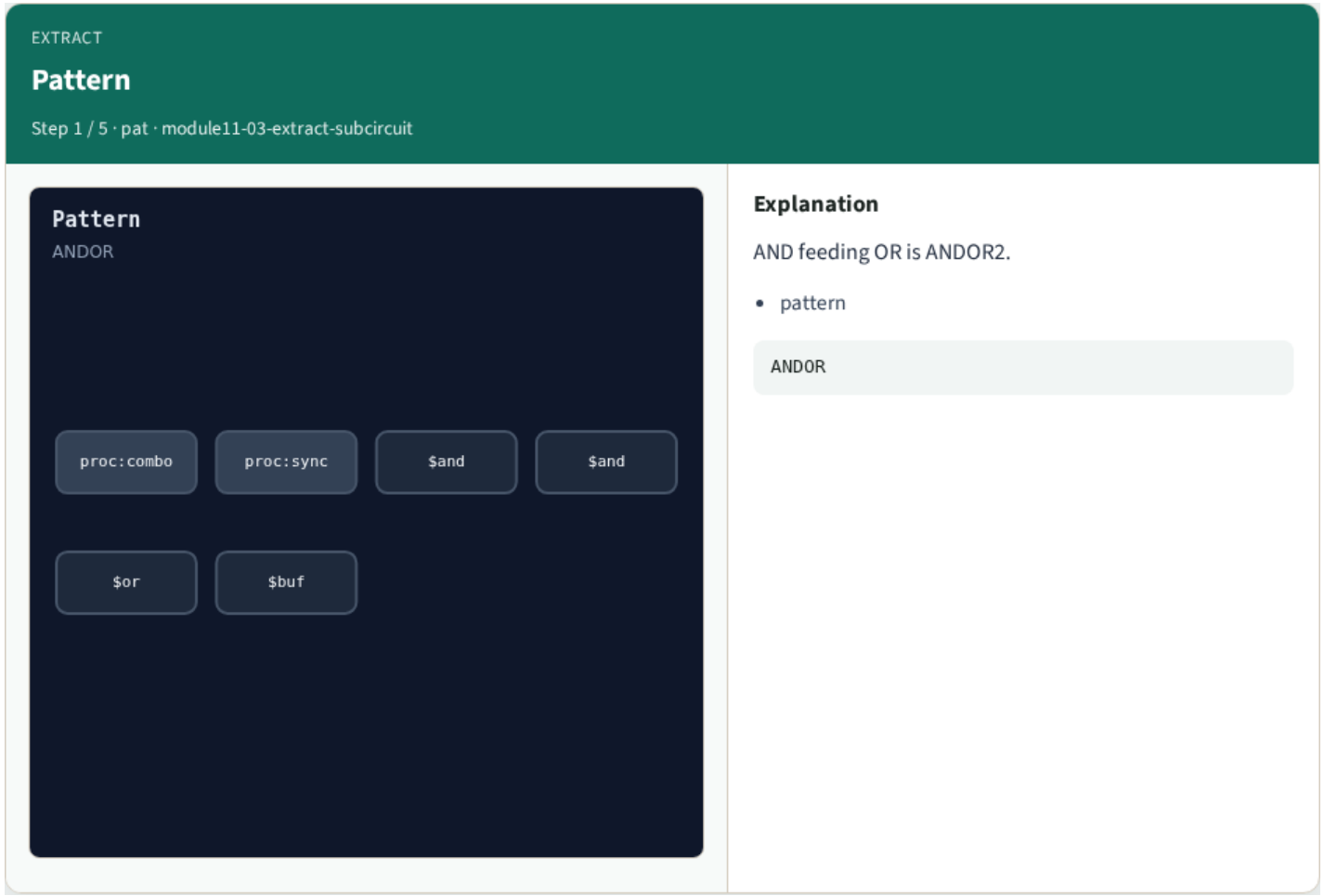
Why extract

Extract spots library-shaped cones and collapses them to a single macro

The idea

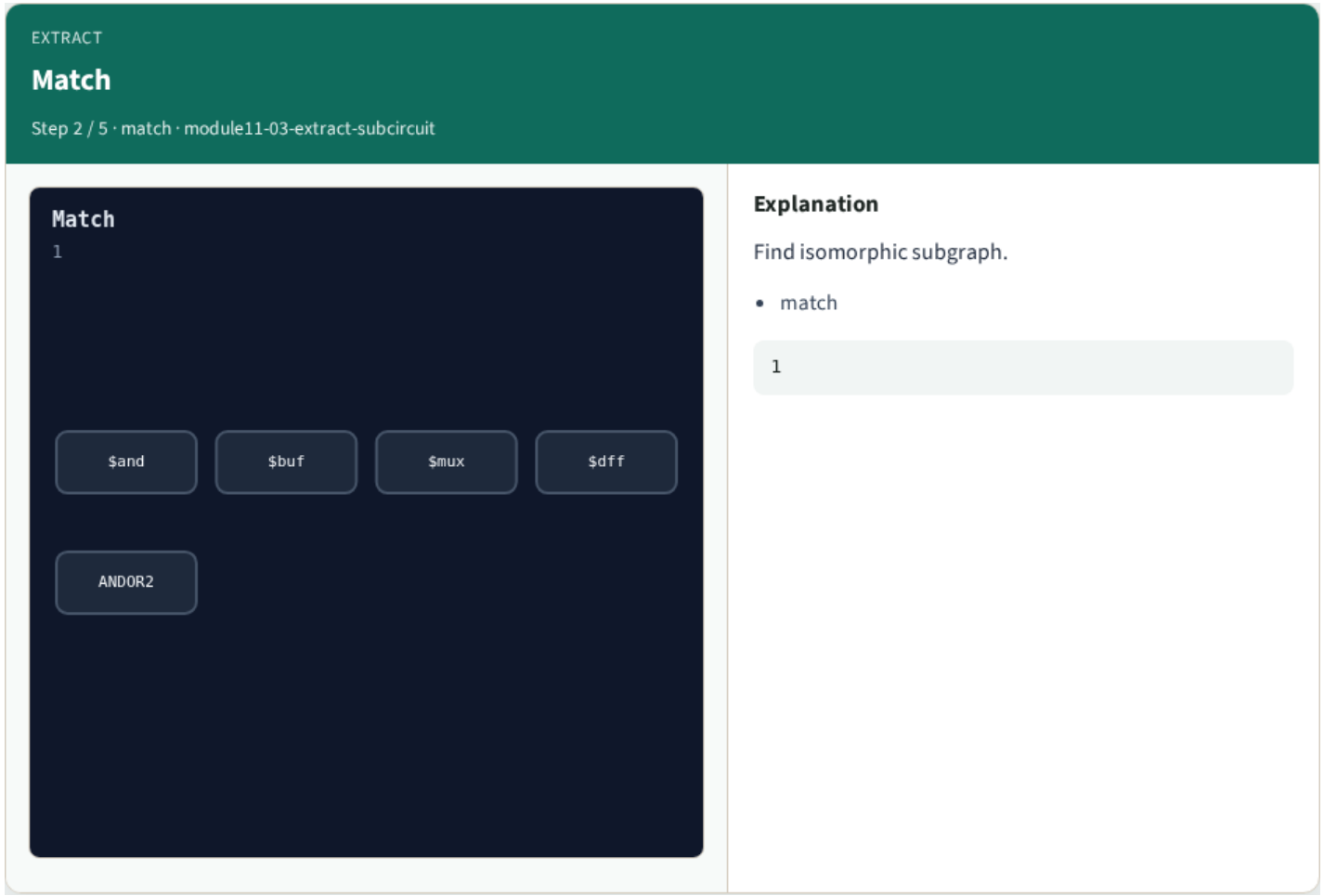
- GOLDEN: matches=1; macro=ANDOR2

Pattern



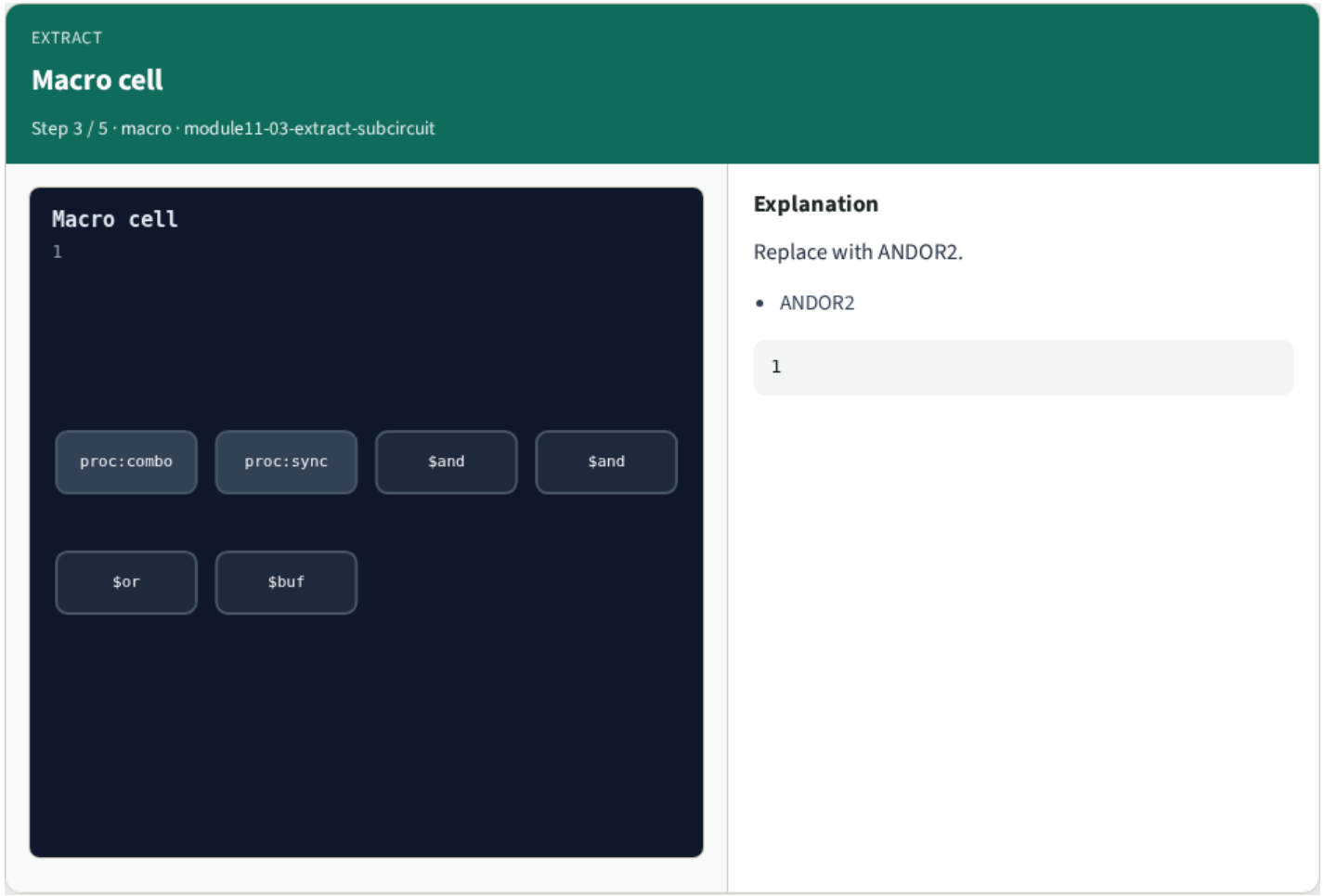
Pattern

Match


EXTRACT
Match
Step 2 / 5 · match · module11-03-extract-subcircuit
Match
1
$and
$buf
$mux
$dff
ANDOR2
Explanation
Find isomorphic subgraph.
match
1

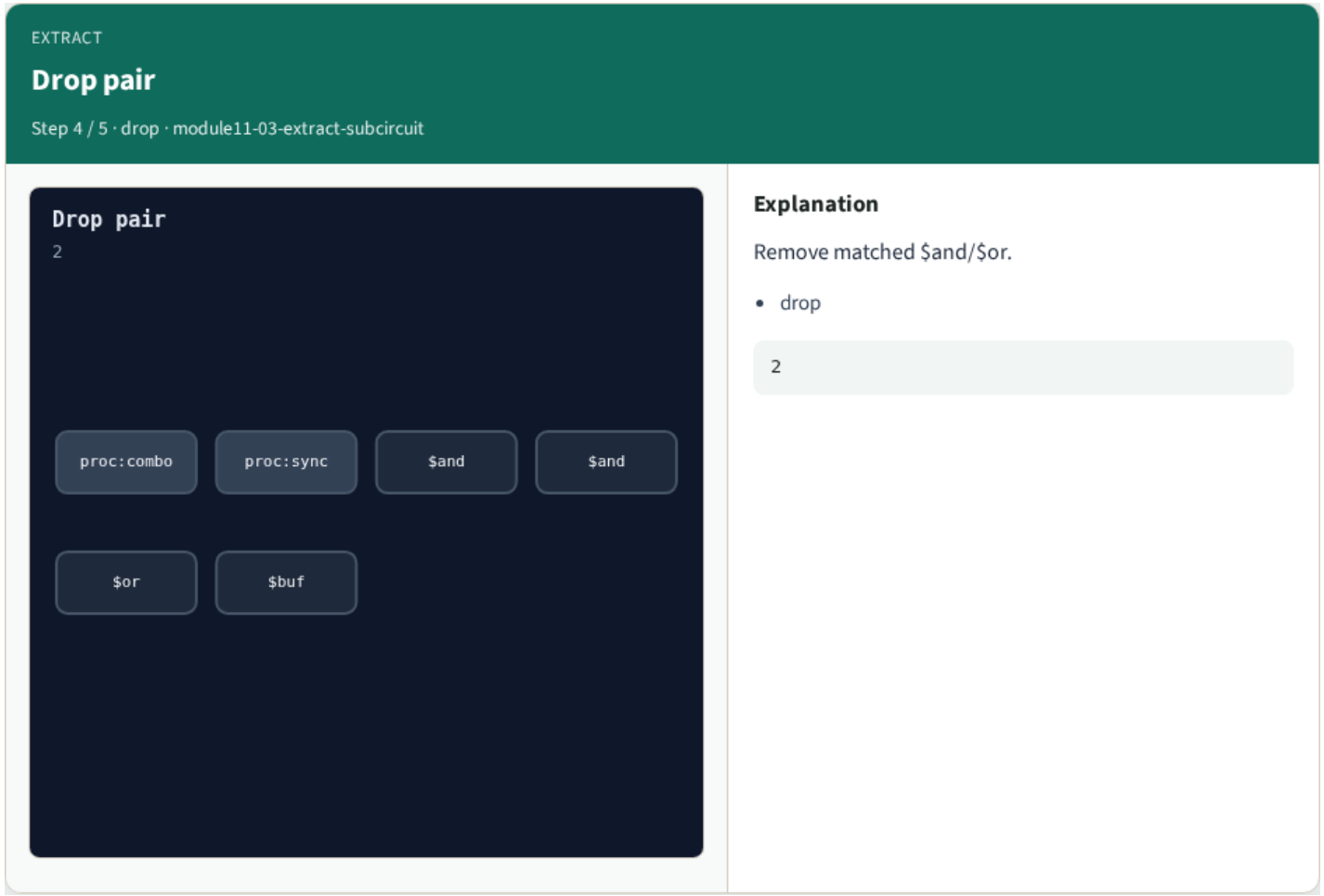
Match

Macro cell



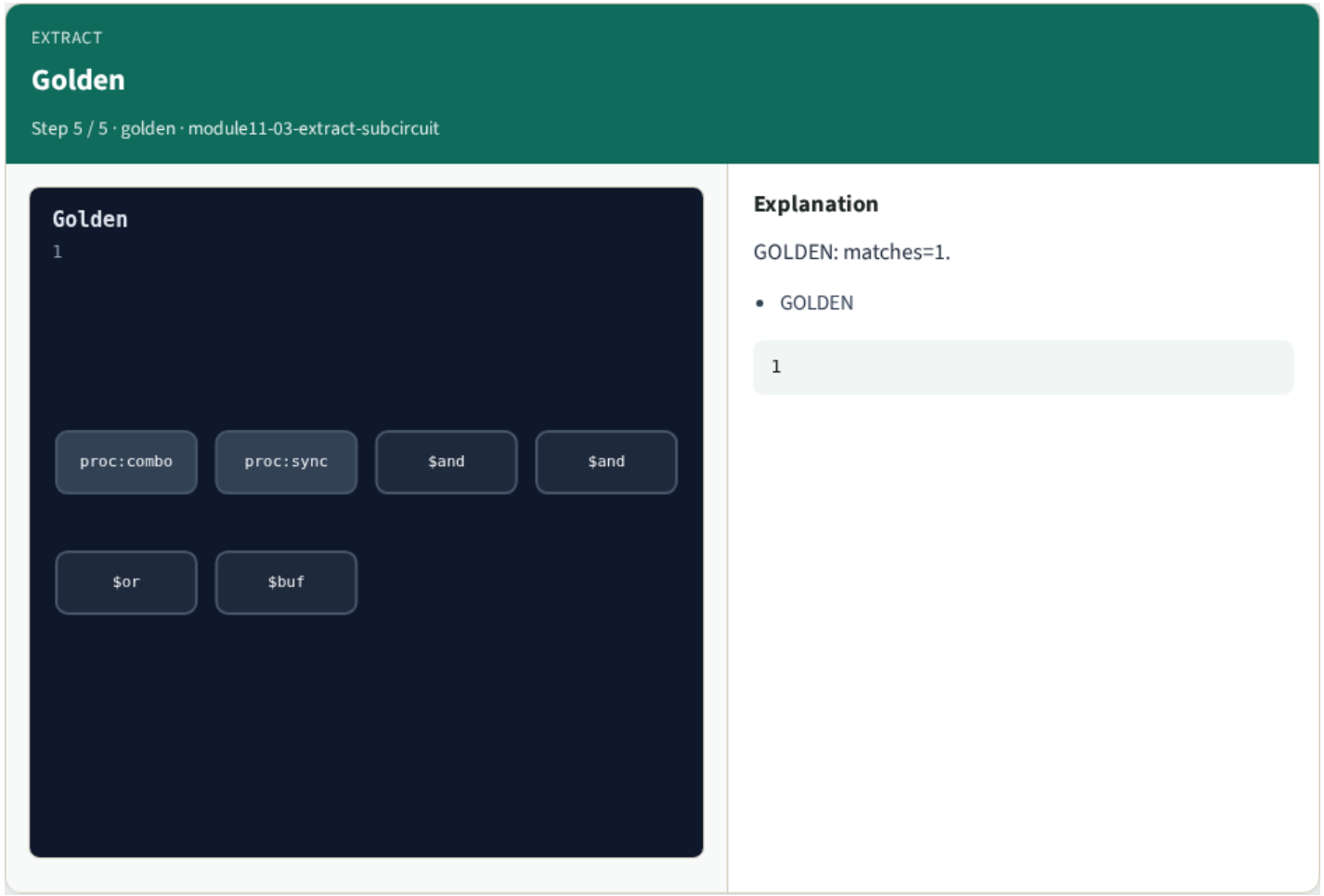
Macro cell

Drop pair



Drop pair

Golden



Golden

Browser lab track

- Open **extract-subcircuit**
- Match the ANDOR2 cone and replace it

Implement track

- Call algo_extract_iso(); assert matches == 1 and macro == 'ANDOR2'

Pitfalls

- Running extract after clean removed the dead OR
- Matching OR without an AND driver

Try these

- From the common folder

Try these — try these

```
cd courses/learn_yosys/common
python3 algo_solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
on post-proc netlist
find dollar or fed by dollar and
replace pair with ANDOR2 macro
ok if matches == 1
```

Next

- Complete the quiz for this part
- Next: memory_map algorithm